

COMPUTER NETWORKS

Week-14

Applications of Hashing

- Mainly used for:
 - integrity checking
 - password storage
 - digital signatures
 - Block chain
- Hashing → Integrity achieved
- Encryption → Confidentiality achieved

Applications of Hashing

- Websites do not store actual passwords.
- Instead:
- Password → Hash
- During login:
- Entered password is hashed again
- Compared with stored hash

- In Block chain technology, Cryptographic hashes link blocks together securely
- Used heavily in Bitcoin and other cryptocurrencies

The Difference..

- Encryption → hides data
- Hashing → verifies data

Hashing Process Flow

Message



Hash Function



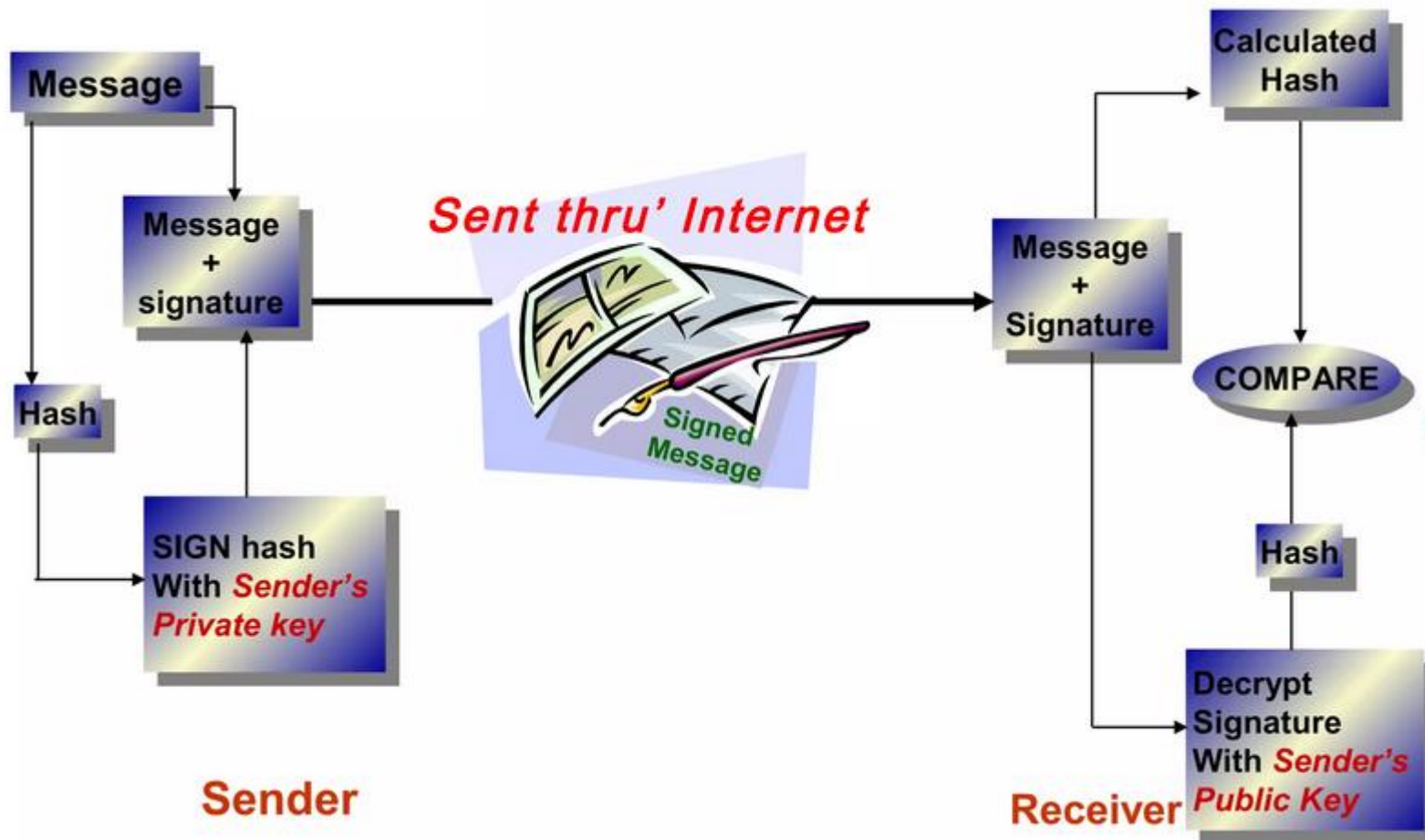
Message Digest



Encrypt digest with PRIVATE key



Digital Signature



Digital Signatures

- NOT a scanned handwritten signature
- It is a mathematical proof created using cryptography

Sender Side

- Creates a hash of the message
- Encrypts that hash using their private key
- The encrypted hash becomes the **digital signature**

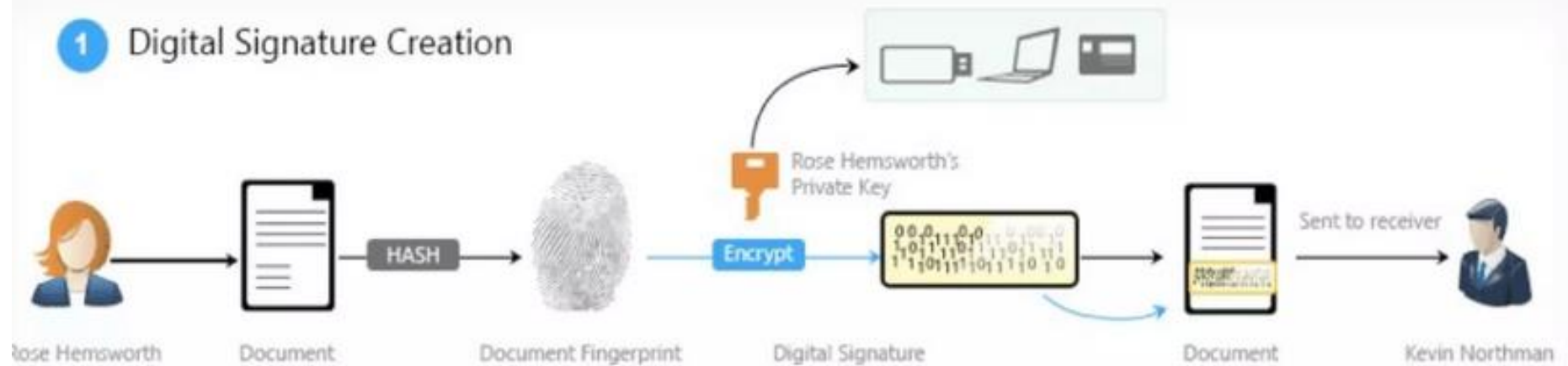
Receiver Side

- Decrypts signature using sender's public key
- Computes hash again from received message
- Compares both hashes
- If both hashes match:
- Message is authentic and was not modified

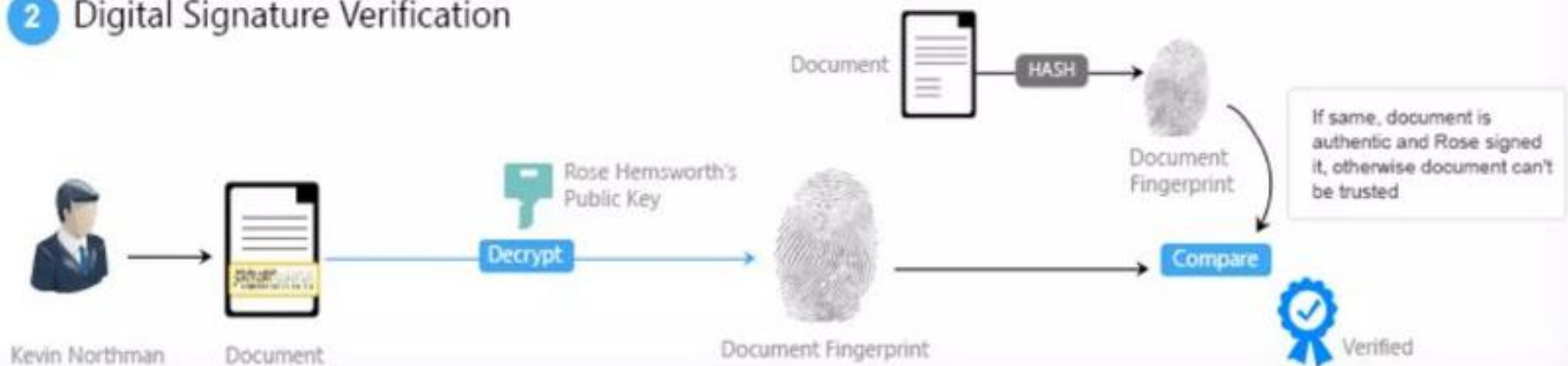
Digital Signatures

- To authenticate the identity of the sender
- To ensure that the original content of a message / a file has been sent unchanged

1 Digital Signature Creation



2 Digital Signature Verification



Keys

- Private Key
 - Only known by the sender or the owner of the data
 - Used to create the digital signature
- Public Key
 - Known to everyone
 - Used to verify the digital signatures

Digital signatures

- Cannot be copied → Authentic
- Depend on the content of the message / File → Integrity
- Can be verified by any computer user and is not error prone

Applications

- E-mails
- Data storage
- Electronic fund transfer
- Software distribution
- Smart cards
- Time stamped signatures

How it works in emails

- **Sender** writes the email
- The email content is passed through a hash function
- A fixed size message digest is generated
- The digest is encrypted using a private key → encrypted digest → digital signature
- The signature is attached to the email

How it works in emails

- Receiver
- Uses the senders public key to decrypt the signature → original digest
- Applies the same hash function (as sender) to the received mail
- Compares the hashes
- If it matches → email is authentic
- Otherwise → email was modified

Examples

- You receiving semester grades through email
- Without digital signatures, an attacker can forge emails to get fake data
- Organizations like Google Gmail, MS outlook etc. use digital signing technology
- In emails, it is like a letter with a unique seal
- Only sender owns the sealing stamp
- Everyone can verify if the seal is genuine
- If someone changes the letter content, the seal breaks mathematically

Digital signature in EFT

- Electronic fund transfer systems
- Transferring money from bank A to bank B
- Customer from bank A creates a transaction
- The banking system applies a hash function to create message digest
- The digest is encrypted using customer's private key → encrypted digest → digital signature
- Transaction is now transmitted
- Bank B gets the transaction details + digital signatures+ customer public key infor.

Digital signature in EFT

- Decryption of signature using public key
- → new hash is generated from received transaction
- Compare both hashes
- If matched → transaction is authentic → sender cannot deny it later → cannot claim later that he/she did not initiate this transaction
- Otherwise transaction is **rejected**
- E.g. SWIFT secure banking communication, PayPal payment verification system, VISA payment system, MasterCard transaction authentication

Digital signature in EFT

- Banks also combine OTPs with digital signatures for multifactor authentication
- Banks also use AI based fraud detection systems